

Mann–Whitney U test for production and test method revisions

Two-sided Mann–Whitney U test comparing the number of revisions of **production** methods versus **test** methods per project. Each row reports the two-sided p -value, Cliff’s δ effect size (d), and its direction sign: + (43 project(s)) indicates that production methods have a higher revision count than test methods ($\delta > 0$); – (65 project(s)) indicates the opposite, i.e. test methods are revised more frequently ($\delta < 0$). Effect size bands follow Romano et al. thresholds for Cliff’s δ : **N** negligible ($|\delta| < 0.147$), **S** small ($0.147 \leq |\delta| < 0.33$), **M** medium ($0.33 \leq |\delta| < 0.474$), **L** large ($|\delta| \geq 0.474$). The pooled all-project comparison reports $p < 0.05$, Cliff’s $\delta = -0.10$, indicating a negligible overall difference with test methods revised more frequently overall.

project	p-value	d-value	<i>Effect size band (Cliff's δ)</i>				
			+/-	N	S	M	L
Activiti	0.00	-0.13	-	x			
ant	0.00	0.38	+			x	
AntennaPod	0.34	0.03	+	x			
Apktool	0.00	-0.21	-		x		
argouml	0.01	0.06	+	x			
async-http-client	0.00	0.71	+				x
atmosphere	0.00	-0.12	-	x			
auto	0.13	-0.04	-	x			
baritone	0.92	-0.02	-	x			
byte-buddy	0.00	0.10	+	x			
caffeine	0.00	-0.40	-			x	
cassandra	0.01	-0.02	-	x			
checkstyle	0.00	-0.38	-			x	
commons-io	0.09	0.03	+	x			
commons-lang	0.00	0.34	+			x	
cucumber-jvm	0.00	-0.32	-		x		
dagger	0.01	0.14	+	x			
dbeaver	0.00	0.16	+		x		
docx4j	0.00	0.21	+		x		
dropwizard	0.00	-0.38	-			x	
dubbo	0.00	-0.33	-		x		
eclipse.jdt.core	0.00	0.12	+	x			
elasticsearch	0.00	-0.22	-		x		
error-prone	0.00	-0.17	-		x		
Essentials	0.95	0.01	+	x			
ExoPlayer	0.00	-0.04	-	x			
facebook-android-sdk	0.01	-0.49	-				x
fastjson	0.00	0.54	+				x
feign	0.00	-0.53	-				x
flink	0.00	-0.34	-			x	
glide	0.44	0.01	+	x			
grpc-java	0.00	-0.17	-		x		
gson	0.03	0.05	+	x			
guava	0.00	0.04	+	x			
guice	0.00	-0.07	-	x			
hadoop	0.00	-0.30	-		x		
hazelcast	0.00	0.03	+	x			
hbase	0.00	-0.26	-		x		
hector	0.03	-0.07	-	x			
hibernate-orm	0.00	-0.07	-	x			
hibernate-search	0.00	-0.58	-				x

project	p-value	d-value	+/-	<i>Effect size band (Cliff's δ)</i>			
				N	S	M	L
HMCL	0.00	-0.19	-		x		
hutool	0.00	-0.14	-	x			
Hystrix	0.00	-0.45	-			x	
iceberg	0.00	-0.23	-		x		
incubator-seata	0.00	-0.04	-	x			
intellij-community	0.00	0.34	+			x	
j2objc	0.00	-0.11	-	x			
javaparser	0.50	0.01	+	x			
jclouds	0.00	-0.20	-		x		
jedis	0.00	-0.12	-	x			
jenkins	0.00	-0.14	-	x			
jetty.project	0.00	-0.12	-	x			
jgit	0.00	0.27	+		x		
jib	0.32	-0.02	-	x			
jmeter	0.00	0.15	+		x		
jna	0.41	-0.02	-	x			
junit4	0.00	0.32	+		x		
junit5	0.00	0.08	+	x			
kafka	0.00	-0.29	-		x		
lombok	0.00	0.38	+			x	
lucene	0.00	-0.15	-	x			
mage	0.00	-0.27	-		x		
metrics	0.00	-0.12	-	x			
Mindustry	0.80	-0.02	-	x			
MinecraftForge	0.00	0.63	+				x
mockito	0.21	-0.02	-	x			
mongo-java-driver	0.94	0.00	+	x			
MPAndroidChart	0.02	0.42	+			x	
mybatis-plus	0.00	0.18	+		x		
nacos	0.06	-0.02	-	x			
netty	0.00	-0.12	-	x			
okhttp	0.16	0.23	+		x		
openmrs-core	0.00	-0.11	-	x			
Paper	0.35	-0.02	-	x			
pentaho-kettle	0.00	0.45	+			x	
plantuml	0.00	0.46	+			x	
pmd	0.00	-0.35	-			x	
presto	0.00	-0.07	-	x			
pulsar	0.00	-0.11	-	x			
redisson	0.00	-0.16	-		x		
rocketmq	0.00	0.10	+	x			
RxJava	0.00	0.25	+		x		
seatunnel	0.00	0.12	+	x			
shardingsphere	0.00	0.06	+	x			
shardingsphere-elasticjob	0.00	-0.37	-			x	
shenyu	0.02	0.03	+	x			
Signal-Server	0.00	-0.29	-		x		
skywalking	0.00	-0.21	-		x		
SmartTube	0.00	0.18	+		x		
solr	0.00	-0.16	-		x		
sonarqube	0.00	-0.30	-		x		

project	p-value	d-value	+/-	<i>Effect size band (Cliff's δ)</i>			
				N	S	M	L
spring-boot	0.00	-0.20	-		x		
spring-cloud-alibaba	0.30	-0.05	-	x			
spring-framework	0.00	-0.22	-		x		
titan	0.00	-0.15	-	x			
tomcat	0.00	0.36	+			x	
vert.x	0.00	-0.19	-		x		
visualvm	0.00	-0.10	-	x			
voldemort	0.00	-0.28	-		x		
vraptor4	0.00	-0.12	-	x			
weka	0.00	0.32	+		x		
wicket	0.00	-0.05	-	x			
wildfly	0.00	0.06	+	x			
wiremock	0.00	0.04	+	x			
xerces2-j	0.00	0.54	+				x
xpipe	0.74	-0.08	-	x			
zxing	0.00	0.15	+		x		
all	0.00	-0.10	-	x			